# Healthcare Data Analytics (PostgreSQL + pgAdmin)

## Overview

A SQL-based data analytics project exploring a synthetic 55,500-record healthcare dataset from Kaggle.
The project covers the full analytics workflow inside PostgreSQL:
schema design, data import, cleaning,
exploratory analysis, aggregation,

time-series analysis, window functions, and reusable views — all executed and verified in pgAdmin.

## Dataset

- **Source:** Healthcare Dataset — Kaggle
- **Size:** 55,500 hospital admission records
- **Fields:** patient demographics (name, age, gender, blood type), medical condition, admission/discharge dates, attending doctor, hospital, insurance provider, billing amount, room number, admission type, medication, and test results.

- **Note:** This is a synthetic dataset generated with Python's Faker library for analytics practice. It contains no real patient data.

## Tools Used

- PostgreSQL (database engine)
- pgAdmin (query execution, CSV import, schema management)

## Project Structure

```
healthcare-sql-analytics/
├── README.md
├── schema.sql
# Table, derived column, and
view definitions
└── queries/
    ├── 01_exploration.sql
    ├── 02_cleaning.sql
```

```
    ├── 03_aggregation.sql
    ├── 04_time_analysis.sql
    ├──
05_window_functions.sql
    └── 06_views.sql
```

## How to Reproduce

1. Create a PostgreSQL database (e.g. `health_analytics`).
2. Run `schema.sql` in pgAdmin's Query Tool to create the `patients` table and its views.
3. Download the CSV from the Kaggle link above.
4. Import it via pgAdmin: right-click the `patients` table → Import/Export Data → Import. Set Header to Yes, and in the

**Columns tab, uncheck `patient_id`** so PostgreSQL auto-generates
it instead of trying to load the `name` column into an integer field.

5. Run the scripts in `/queries` in numeric order.

## Key Findings

*(Fill in with your actual results from the queries you ran)*

- The most common medical condition is **___**, accounting for **___** cases.
- The most expensive condition to treat on average is **___**, at **$___** per admission.

- Hospital ___ has the highest patient volume; Hospital ___ has the longest average stay.
- Admissions peak in ___ (month) / on ___ (day of week).
- Insurance provider ___ covers the largest share of patients.
- A small number of records (see `02_cleaning.sql`) share identical name/date/hospital combinations — a known characteristic of this synthetic dataset rather than an import error.

## Skills Demonstrated

- Relational schema design (`CREATE TABLE`, data types, primary keys)
- Bulk data import via `COPY` / pgAdmin's Import/Export tool
- Data cleaning: text normalization, null/duplicate checks, integrity validation
- Aggregation with `GROUP BY`, `HAVING`, and multiple aggregate functions
- Date/time analysis with `DATE_TRUNC`, `EXTRACT`, and `TO_CHAR`
- Window functions: `RANK() OVER (PARTITION BY ...)`, running totals, percentage-of-total

- Views for a reusable, dashboard-ready data layer

## Author

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